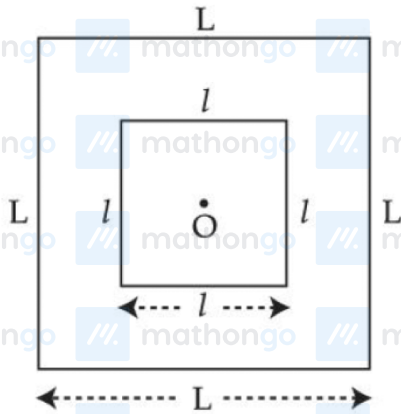


Q1 - 25 July - Shift 1

A small square loop of wire of side l is placed inside a large square loop of wire L ($L \gg l$). Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is :



- (A) $\frac{2\sqrt{2}\mu_0 L^2}{\pi l}$ (B) $\frac{\mu_0 l^2}{2\sqrt{2}\pi L}$
 (C) $\frac{2\sqrt{2}\mu_0 l^2}{\pi L}$ (D) $\frac{\mu_0 L^2}{2\sqrt{2}\pi l}$

Space for your notes:

Q2 - 25 July - Shift 2

Magnetic flux (in weber) in a closed circuit of resistance $20\ \Omega$ varies with time $t(s)$ as $\phi = 8t^2 - 9t + 5$. The magnitude of the induced current at $t = 0.25\ s$ will be _____ mA

Space for your notes:

Q3 - 26 July - Shift 2

Questions

MathonGo

A velocity selector consists of electric field $\vec{E} = E\hat{k}$ and magnetic field $\vec{B} = B\hat{j}$ with $B = 12 \text{ mT}$. The value E required for an electron of energy 728 eV moving along the positive x -axis to pass undeflected is :

(Given, mass of electron $= 9.1 \times 10^{-31} \text{ kg}$)

(A) 192 kVm^{-1} (B) 192 mVm^{-1}

(C) 9600 kVm^{-1} (D) 16 kVm^{-1}

Space for your notes:

Q4 - 27 July - Shift 1

A beam of light travelling along X -axis is described by the electric field $E_y = 900 \sin \omega(t - x/c)$. The ratio of electric force to magnetic force on a charge q moving along Y -axis with a speed of $3 \times 10^7 \text{ ms}^{-1}$ will be :

[Given speed of light $= 3 \times 10^8 \text{ ms}^{-1}$]

(A) $1 : 1$ (B) $1 : 10$

(C) $10 : 1$ (D) $1 : 2$

Space for your notes:

Q5 - 27 July - Shift 2

A conducting circular loop is placed in $X - Y$ plane in presence of magnetic field $\vec{B} = (3t^3\hat{j} + 3t^2\hat{k})$ in SI unit. If the radius of the loop is 1 m , the induced emf in the loop, at time, $t = 2 \text{ s}$ is $n\pi \text{ V}$. The value of n is.....

Space for your notes:

Q6 - 28 July - Shift 2

#MathBoleTohMathonGo

A transformer operating at primary voltage 8 kV and secondary voltage 160 V serves a load of 80 kW. Assuming the transformer to be ideal with purely resistive load and working on unity power factor, the loads in the primary and secondary circuit would be

- (A) $800\ \Omega$ and $1.06\ \Omega$ (B) $10\ \Omega$ and $500\ \Omega$
(C) $800\ \Omega$ and $0.32\ \Omega$ (D) $1.06\ \Omega$ and $500\ \Omega$

Space for your notes:

Answer Key

Q1 (C)

Q2 (250)

Q3 (A)

Q4 (C)

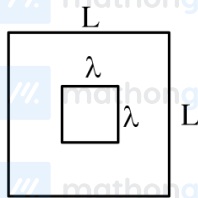
Q5 (12)

Q6 (C)

#MathBoleTohMathonGo

Q1 (C)Assuming current I in outer loop magnetic field at

$$\text{centre} = 4 \times \frac{\mu_0 i}{4\pi \times \frac{L}{2}} \times (2 \sin 45^\circ) = \frac{2\sqrt{2}\mu_0 i}{\pi L}$$



$$M = \frac{\text{Flux through inner loop}}{i}$$

$$M = \frac{2\sqrt{2}\mu_0 \ell^2}{\pi L}$$

Q2 (250)

$$\phi = 8t^2 - 9t + 5$$

$$\text{emf} = -\frac{d\phi}{dt} = -(16t - 9)$$

At $t = 0.25$ s

$$\text{Emf} = -[(16 \times 0.25) - 9] = 5V$$

$$\text{Current} = \frac{\text{Emf}}{\text{Resistance}} = \frac{5V}{20\Omega}$$

$$= \frac{1}{4} A = \frac{1000}{4} \text{mA} = 250 \text{mA}$$

#MathBoleTohMathonGo

Q3 (A)

$$\vec{E} = E \hat{k} \quad B = 12 \text{ mT}$$

$$\vec{B} = B \hat{j} \quad \text{Energy} = 728 \text{ eV}$$

$$\text{Energy} = \frac{1}{2}mv^2$$

$$728 \text{ eV} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$$

$$728 \times 1.6 \times 10^{-19} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$$

$$v = 16 \times 10^6 \text{ m/s}$$

$$E = vB$$

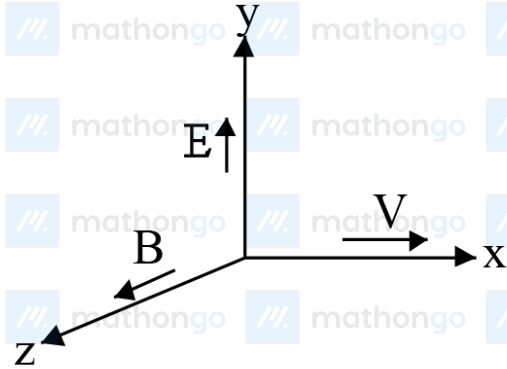
$$E = 16 \times 10^6 \times 12 \times 10^{-3}$$

$$E = 192 \times 10^3 \text{ V/m}$$

Q4 (C)

$$E_y = 900 \sin\left(\omega t - \frac{\omega x}{c}\right)$$

$$E_0 = 900$$



$$F_E = qE_0$$

$$F_B = qvB_0$$

$$\frac{F_E}{F_B} = \frac{E_0}{vB_0} = \frac{c}{v} = \frac{3 \times 10^8}{3 \times 10^7} = 10:1$$

Q5 (12)

$$\phi = \vec{B} \cdot \vec{A}$$

$$= (3t^3 \hat{j} + 3t^2 \hat{k}) \cdot (\pi(1)^2 \hat{k})$$

$$\phi = 3t^2 \pi$$

$$\varepsilon_{\text{IND}} = \left| \frac{d\phi}{dt} \right| = 6t\pi$$

$$\text{at } t = 2, \varepsilon_{\text{IND}} = 12$$

Q6 (C)

$$\frac{(8 \times 10^3)^2}{R_P} = 80 \times 10^3$$

$$R_P = 800 \Omega$$

$$\frac{(160)^2}{R_S} = 80 \times 10^3$$

$$R_S = 0.32 \Omega$$